25 Maths CT-1 solved

Set 1

CO1

a. Define axiomatic definition of probability. (2 marks)

Let S be a sample space. A probability measure P is a function that assigns a real number $P(A)$ to each event A in S, satisfying the following three Kolmogorov's axioms:

1. **Non-negativity:** For any event A, the probability is non-negative: $P(A) \geq 0$.
2. **Normalization:** The probability of the entire sample space is 1: $P(S) = 1$.
3. **Additivity:** For any sequence of mutually exclusive (disjoint) events $A_1, A_2, A_3, \ldots$ (where $A_i \cap A_j = \emptyset$ for $i \neq j$), the probability of their union is the sum of their probabilities:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

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b. In a bolt factory machine M_1, M_2 and M_3 manufacture respectively 25%, 35% and 40% of the total of their output with 5%, 4%, and 2% respectively are defective bolts. One bolt is chosen at random from the product and it is found defective. What is probability that it is manufactured in machine M_2? (5 marks)

Let's define the events:

- E_1: Bolt is produced by machine M_1
- E_2: Bolt is produced by machine M_2

- E_3: Bolt is produced by machine M_3
- D: Bolt is defective

Here is a structured table of the given probabilities:

Machine	**Prior Probability** $P(E_i)$	**Conditional Prob of Defect** $P(D\|E_i)$	**Joint Probability** $P(E_i \cap D) = P(E_i) \times P(D\|E_i$
M_1	$P(E_1) = 0.25$	$P(D\|E_1) = 0.05$	$0.25 \times 0.05 = 0.0125$
M_2	$P(E_2) = 0.35$	$P(D\|E_2) = 0.04$	$0.35 \times 0.04 = 0.0140$
M_3	$P(E_3) = 0.40$	$P(D\|E_3) = 0.02$	$0.40 \times 0.02 = 0.0080$
Total	$\sum P(E_i) = 1.00$	-	$P(D) = 0.0345$

We need to find $P(E_2|D)$, which is the probability that the bolt came from M_2 given that it is defective. Using Bayes' Theorem:

$$P(E_2|D) = \frac{P(E_2 \cap D)}{P(D)} = \frac{P(E_2)P(D|E_2)}{P(E_1)P(D|E_1) + P(E_2)P(D|E_2) + P(E_3)}$$

Substitute the values from our table:

$$P(E_2|D) = \frac{0.0140}{0.0125 + 0.0140 + 0.0080}$$

$$P(E_2|D) = \frac{0.0140}{0.0345} = \frac{140}{345} = \frac{28}{69} \approx 0.4058$$

Answer: The probability that it is manufactured in machine M_2 is $\frac{28}{69}$ (or roughly **40.58%**).

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c. A random variable X has the pdf: $f(x) = cx^2$ for $0 \le x \le 1$ and 0 otherwise. Find the value of c and $P(X > 3/4)$. (3 marks)

Part 1: Find the value of c

For $f(x)$ to be a valid Probability Density Function (PDF), the total area under the curve must be 1.

$$\int_{-\infty}^{\infty} f(x)dx = 1 \Rightarrow \int_0^1 cx^2 dx = 1$$

$$c\left[\frac{x^3}{3}\right]_0^1 = 1$$

$$c\left(\frac{1^3}{3} - \frac{0^3}{3}\right) = 1 \Rightarrow \frac{c}{3} = 1 \Rightarrow \mathbf{c = 3}$$

So, $f(x) = 3x^2$ for $0 \le x \le 1$.

Part 2: Find $P(X > 3/4)$

$$P\left(X > \frac{3}{4}\right) = \int_{3/4}^1 f(x)dx = \int_{3/4}^1 3x^2 dx$$

$$= \left[x^3\right]_{3/4}^1 = 1^3 - \left(\frac{3}{4}\right)^3 = 1 - \frac{27}{64} = \mathbf{\frac{37}{64}}$$

(Which is approximately **0.5781**)

CO2

a. Derive the expressions of mean and variance of Binomial distribution with given parameters. (3+3 marks)

Let $X \sim Binomial(n, p)$. The Probability Mass Function (PMF) is:

$P(X = x) = \binom{n}{x} p^x q^{n-x}$ for $x = 0, 1, 2, \ldots, n$, where $q = 1 - p$.

Mean (μ):

$$\mu = E(X) = \sum_{x=0}^{n} x \cdot P(X = x) = \sum_{x=0}^{n} x \binom{n}{x} p^x q^{n-x}$$

Since the term is 0 when $x = 0$, we start the sum at $x = 1$:

$$E(X) = \sum_{x=1}^{n} x \frac{n!}{x!(n-x)!} p^x q^{n-x} = \sum_{x=1}^{n} n \frac{(n-1)!}{(x-1)!(n-x)!} p \cdot p^{x-1} q^{n-x}$$

Factor out np:

$$E(X) = np \sum_{x=1}^{n} \binom{n-1}{x-1} p^{x-1} q^{(n-1)-(x-1)}$$

Using the Binomial theorem, the summation is the expansion of $(p+q)^{n-1}$.

$$E(X) = np(p+q)^{n-1}$$

Since $p + q = 1$, **Mean** $= np$.

Variance (σ^2):

$\sigma^2 = E(X^2) - [E(X)]^2$. Let's find $E(X(X-1))$ first to make finding $E(X^2)$ easier.

$$E(X(X-1)) = \sum_{x=0}^{n} x(x-1) \binom{n}{x} p^x q^{n-x}$$

Starting from $x = 2$:

$$E(X(X-1)) = \sum_{x=2}^{n} x(x-1) \frac{n!}{x(x-1)(x-2)!(n-x)!} p^2 p^{x-2} q^{n-x}$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \binom{n-2}{x-2} p^{x-2} q^{(n-2)-(x-2)}$$

Similar to before, the summation evaluates to $(p+q)^{n-2} = 1$.

$$E(X(X-1)) = n(n-1)p^2 = n^2p^2 - np^2$$

Now, substitute back to find variance:
$E(X^2) = E(X(X-1)) + E(X) = n^2p^2 - np^2 + np$
$\sigma^2 = E(X^2) - [E(X)]^2 = (n^2p^2 - np^2 + np) - (np)^2$
$\sigma^2 = n^2p^2 - np^2 + np - n^2p^2 = np - np^2 = np(1-p) = \mathbf{npq}$.

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b. The probability that a Poisson variate X takes +ve value is $(1 - e^{-2})$. Find variance and probability that X lies between -1 and 1.5. (3 marks)

The PMF of a Poisson distribution is $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ for $x = 0, 1, 2, \ldots$
A "positive value" means $X \geq 1$.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{e^{-\lambda}\lambda^0}{0!} = 1 - e^{-\lambda}$$

Given $P(X \geq 1) = 1 - e^{-2}$, we can equate:

$$1 - e^{-\lambda} = 1 - e^{-2} \Rightarrow \lambda = 2$$

For a Poisson distribution, the **Variance** $= \lambda = \mathbf{2}$.
To find the probability that X lies between -1 and 1.5:
We need $P(-1 < X < 1.5)$. Since Poisson variables only take non-negative integer values $(0, 1, 2 \ldots)$, the only valid values for X in this range are 0 and 1.

$$P(-1 < X < 1.5) = P(X = 0) + P(X = 1)$$

$$= \frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} = e^{-2} + 2e^{-2} = \mathbf{3e^{-2}}$$

(Which is approximately $3 \times 0.1353 = \mathbf{0.406}$)

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c. In a sample of 120 workers in a factory, the mean and s.d. of wages were INR 11.35 and INR 3.03 respectively. Find the percentage of workers getting wages between INR 9 - INR 17 in the whole factory, assuming that the wages are normally distributed. (Given: area under standard normal curve from 0 - 0.78 is 0.2823 and 0 - 1.86 is 0.4686). (3 marks)

Let X be the wage. $X \sim N(\mu = 11.35, \sigma = 3.03)$. We need $P(9 < X < 17)$.

First, we standardize the values using $Z = \frac{X-\mu}{\sigma}$:

For $X_1 = 9$:

$$z_1 = \frac{9 - 11.35}{3.03} = \frac{-2.35}{3.03} \approx -0.7755 \approx -0.78$$

For $X_2 = 17$:

$$z_2 = \frac{17 - 11.35}{3.03} = \frac{5.65}{3.03} \approx 1.8646 \approx 1.86$$

Now we find the area:

$$P(9 < X < 17) = P(-0.78 < Z < 1.86)$$

By symmetry of the normal curve, $P(-0.78 < Z < 0) = P(0 < Z < 0.78)$.

$$P(-0.78 < Z < 1.86) = P(0 < Z < 0.78) + P(0 < Z < 1.86)$$

Using the given areas:

$$P = 0.2823 + 0.4686 = 0.7509$$

To get the percentage, multiply by 100: $\mathbf{75.09\%}$.

CO3

a. Two lines of regression are given by $x + 2y = 5$ and $2x + 3y = 8$ and $\sigma_x^2 = 12$. Calculate mean of x, mean of y, σ_y, and r. (5 marks)

1. Mean of x ($\bar{x}$) and mean of y ($\bar{y}$)

The regression lines always intersect at the point $(\bar{x}, \bar{y})$. Therefore, we solve the two equations simultaneously:

Equation 1: $x + 2y = 5 \Rightarrow x = 5 - 2y$

Equation 2: $2x + 3y = 8$

Substitute (1) into (2):

$2(5 - 2y) + 3y = 8 \Rightarrow 10 - 4y + 3y = 8 \Rightarrow -y = -2 \Rightarrow \bar{\mathbf{y}} = \mathbf{2}$

Substitute $y = 2$ back into (1):

$x + 2(2) = 5 \Rightarrow x + 4 = 5 \Rightarrow \bar{\mathbf{x}} = \mathbf{1}$

2. Finding the correlation coefficient (r)

Let's make an assumption about which equation is Y on X and which is X on Y.

Let Eq 1 ($x + 2y = 5$) be the regression line of Y on X:

$$2y = -x + 5 \Rightarrow y = -0.5x + 2.5$$

Here, the regression coefficient $b_{yx} = -0.5$

Let Eq 2 ($2x + 3y = 8$) be the regression line of X on Y:

$$2x = -3y + 8 \Rightarrow x = -1.5y + 4$$

Here, the regression coefficient $b_{xy} = -1.5$

Property check: $r^2 = b_{yx} \cdot b_{xy} = (-0.5)(-1.5) = 0.75$.

Since $r^2 \leq 1$, our assumption is valid.

Because both regression coefficients are negative, r must also be negative.

$$\mathbf{r} = -\sqrt{0.75} = -\frac{\sqrt{3}}{2} \approx \mathbf{-0.866}$$

3. Finding Standard Deviation of y (σ_y)

Given $\sigma_x^2 = 12 \Rightarrow \sigma_x = \sqrt{12} = 2\sqrt{3}$.

We use the formula for the regression coefficient $b_{yx} = r\frac{\sigma_y}{\sigma_x}$:

$$-0.5 = \left(-\frac{\sqrt{3}}{2}\right)\frac{\sigma_y}{2\sqrt{3}}$$

$$-0.5 = -\frac{\sigma_y}{4} \Rightarrow \boldsymbol{\sigma_y = 2}$$

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b. If m_2' and m_2 are respectively the second moment about an arbitrary origin "a" and that about mean of x, then show that $m_2' = m_2 + d^2$, where $d = (\text{mean of } x) - a$. (3 marks)

Let the random variable be X, and its mean be $\bar{x} = E(X)$.

By definition:

- Second moment about mean (which is Variance): $m_2 = E[(X - \bar{x})^2]$
- Second moment about arbitrary origin a: $m_2' = E[(X - a)^2]$

We are given $d = \bar{x} - a$, which rearranges to $a = \bar{x} - d$.

Substitute a into the expression for m_2':

$$m_2' = E[(X - (\bar{x} - d))^2] = E[((X - \bar{x}) + d)^2]$$

Expand the square:

$$m_2' = E[(X - \bar{x})^2 + 2d(X - \bar{x}) + d^2]$$

Apply the linearity of expectation ($E[A + B] = E[A] + E[B]$):

$$m_2' = E[(X - \bar{x})^2] + 2d \cdot E[X - \bar{x}] + E[d^2]$$

Now, evaluate each term:

1. $E[(X - \bar{x})^2]$ is exactly our definition of m_2.
2. $E[X - \bar{x}] = E[X] - E[\bar{x}] = \bar{x} - \bar{x} = 0$. So the middle term becomes 0.

3. Since d is a constant, $E[d^2] = d^2$.

Putting it all together:

$$m'_2 = m_2 + 2d(0) + d^2$$

$m'_2 = m_2 + d^2$ **(Hence shown)**

Set 2

CO1

a. Define mutually exhaustive events. If A and B are two independent events either one is null then $P(A \cap B) =$? (2 marks)

1. Definition of Mutually Exhaustive Events:

A set of events is considered **mutually exhaustive** (or simply exhaustive) if at least one of the events must occur every time the experiment is performed. Mathematically, a set of events $E_1, E_2, \ldots, E_n$ is exhaustive if their union is equal to the entire sample space S.

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) = P(S) = 1$$

2. Finding $P(A \cap B)$:

- Because A and B are independent events, the probability of their intersection is the product of their individual probabilities: $P(A \cap B) = P(A)P(B)$.
- We are given that "either one is null". A null event (an impossible event) has a probability of 0. So, either $P(A) = 0$ or $P(B) = 0$.
- Therefore, $P(A \cap B) = P(A) \times P(B) = 0$. **Answer:** $P(A \cap B) = \mathbf{0}$.

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b. A bag contains 4 R and 3 B balls. Two drawings of 2 balls are made. What is the probability of drawing first 2 R balls and second 2 B balls, i) if the balls

are returned to the bag after the first draw and ii) if the balls are not returned after the first draw? (5 marks)

Let's define the events:

- E_1: Drawing 2 Red (R) balls in the first draw.
- E_2: Drawing 2 Black (B) balls in the second draw. Initial bag state: 4 Red + 3 Black = 7 balls total.

Draw Type	**Probability of Event 1 (P(E1))**	**Probability of Event 2 (P(E2 |E1))**	**Joint Probability P(E1∩E2)**
Case i: With Replacement	$\frac{\binom{4}{2}}{\binom{7}{2}} = \frac{6}{21} = \frac{2}{7}$	(Bag resets to 7 balls) $\frac{\binom{3}{2}}{\binom{7}{2}} = \frac{3}{21} = \frac{1}{7}$	$\frac{2}{7} \times \frac{1}{7} = \mathbf{\frac{2}{49}}$
Case ii: Without Replacement	$\frac{\binom{4}{2}}{\binom{7}{2}} = \frac{6}{21} = \frac{2}{7}$	(Bag now has 2 R, 3 B = 5 balls) $\frac{\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10}$	$\frac{2}{7} \times \frac{3}{10} = \frac{6}{70} = \mathbf{\frac{3}{35}}$

Answers:

i) With replacement: $\frac{2}{49}$

ii) Without replacement: $\frac{3}{35}$

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c. Show that $f(x) = x$ when $0 \le x \le 1$, $k - x$ when $1 < x \le 2$, and 0 otherwise, is a pdf for a suitable value of k. Hence find $P(X \le 1.5)$. (3

marks)

Part 1: Find the value of k

For $f(x)$ to be a valid Probability Density Function (PDF), the total area under the curve must be 1:

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Break the integral into the given intervals:

$$\int_0^1 x\,dx + \int_1^2 (k-x)\,dx = 1$$

Evaluate the integrals:

$$\left[\frac{x^2}{2}\right]_0^1 + \left[kx - \frac{x^2}{2}\right]_1^2 = 1$$

$$\left(\frac{1}{2} - 0\right) + \left[\left(2k - \frac{4}{2}\right) - \left(k - \frac{1}{2}\right)\right] = 1$$

$$\frac{1}{2} + 2k - 2 - k + \frac{1}{2} = 1$$

$$k - 1 = 1 \implies \mathbf{k = 2}$$

The function is $f(x) = 2 - x$ for $1 < x \leq 2$. (Note: $f(x) \geq 0$ for all valid ranges, satisfying the other PDF condition).

Part 2: Find $P(X \leq 1.5)$

$$P(X \leq 1.5) = \int_0^{1.5} f(x)dx = \int_0^1 x\,dx + \int_1^{1.5} (2-x)\,dx$$

$$= \left[\frac{x^2}{2}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^{1.5}$$

$$= \frac{1}{2} + \left[\left(2(1.5) - \frac{1.5^2}{2}\right) - \left(2(1) - \frac{1^2}{2}\right)\right]$$

$$= 0.5 + [(3 - 1.125) - (2 - 0.5)]$$

$$= 0.5 + [1.875 - 1.5] = 0.5 + 0.375 = \mathbf{0.875}$$

*(Which is exactly $\frac{7}{8}$)**

CO2

a. Derive the expressions of mean and variance of Poisson distribution with given parameter. (3+3 marks)

Let $X \sim Poisson(\lambda)$. The PMF is $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ for $x = 0, 1, 2, \ldots$

Mean (μ):

$$\mu = E(X) = \sum_{x=0}^{\infty} x \cdot P(X = x) = \sum_{x=1}^{\infty} x \frac{e^{-\lambda}\lambda^x}{x!}$$

(We start at $x = 1$ because the term is 0 when $x = 0$)

$$E(X) = \sum_{x=1}^{\infty} \frac{e^{-\lambda}\lambda \cdot \lambda^{x-1}}{(x-1)!} = \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

Let $y = x - 1$. As x goes from 1 to ∞, y goes from 0 to ∞:

$$E(X) = \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!}$$

Since the Taylor series expansion for e^{λ} is $\sum_{y=0}^{\infty} \frac{\lambda^y}{y!}$, we get:

$$E(X) = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda$$

Variance (σ^2):

To find variance, we first find $E(X(X-1))$:

$$E(X(X-1)) = \sum_{x=0}^{\infty} x(x-1)\frac{e^{-\lambda}\lambda^x}{x!} = \sum_{x=2}^{\infty} \frac{e^{-\lambda}\lambda^2 \cdot \lambda^{x-2}}{(x-2)!}$$

Pulling out constants:

$$E(X(X-1)) = \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} = \lambda^2 e^{-\lambda} \cdot e^{\lambda} = \lambda^2$$

Now, substitute this into the variance formula $\sigma^2 = E(X^2) - [E(X)]^2$:

$$E(X^2) = E(X(X-1)) + E(X) = \lambda^2 + \lambda$$

$$\sigma^2 = (\lambda^2 + \lambda) - (\lambda)^2 = \lambda$$

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b. The overall percentage of failures in a certain examination is 40. What is the probability that out of 6 candidates at least 4 passed the exam? (3 marks)

Let success be passing the exam.

- Probability of failure $q = 0.40$
- Probability of passing $p = 1 - 0.40 = 0.60$
- Number of trials $n = 6$ Let X be the number of passing students. $X \sim Binomial(n = 6, p = 0.6)$. We need to find $P(X \geq 4) = P(X = 4) + P(X = 5) + P(X = 6)$. Using the Binomial PMF formula $P(X = x) = \binom{n}{x} p^x q^{n-x}$:

Event	Formula	Calculation	Value
$P(X = 4)$	$\binom{6}{4}(0.6)^4(0.4)^2$	$15 \times 0.1296 \times 0.16$	0.31104
$P(X = 5)$	$\binom{6}{5}(0.6)^5(0.4)^1$	$6 \times 0.07776 \times 0.4$	0.186624
$P(X = 6)$	$\binom{6}{6}(0.6)^6(0.4)^0$	$1 \times 0.046656 \times 1$	0.046656

Total Probability = $0.31104 + 0.186624 + 0.046656 = \mathbf{0.54432}$ (or roughly **54.43%**).

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c. The number of death per day due to road accidents and due to other causes independently follow Poisson distribution with parameter 2 and 6 respectively. Find the probability that the total number of deaths on a particular day is 2 or fewer (Given: $e^{-8} = 0.0003$). (3 marks)

Let X = deaths due to road accidents $\sim Poisson(\lambda_1 = 2)$.

Let Y = deaths due to other causes $\sim Poisson(\lambda_2 = 6)$.

Because X and Y are independent Poisson variables, their sum $Z = X + Y$ (total deaths) also follows a Poisson distribution where the parameter is the sum of the individual parameters:

$$\lambda_Z = \lambda_1 + \lambda_2 = 2 + 6 = 8$$

So, $Z \sim Poisson(8)$. We need to find $P(Z \leq 2)$:

$$P(Z \leq 2) = P(Z = 0) + P(Z = 1) + P(Z = 2)$$

$$P(Z \leq 2) = \frac{e^{-8}8^0}{0!} + \frac{e^{-8}8^1}{1!} + \frac{e^{-8}8^2}{2!}$$

$$P(Z \leq 2) = e^{-8}\left(1 + 8 + \frac{64}{2}\right) = e^{-8}(1 + 8 + 32) = 41e^{-8}$$

Using the given value $e^{-8} = 0.0003$:

$$P(Z \leq 2) = 41 \times 0.0003 = \mathbf{0.0123}$$

CO3

a. Two variates have the least squares regression lines $x + 4y + 3 = 0$ and $4x + 9y + 5 = 0$. Find their mean values and correlation coefficient. (5 marks)

1. Mean values ($\bar{x}$ and $\bar{y}$)

The regression lines always intersect at their means $(\bar{x}, \bar{y})$. Therefore, we solve the two equations simultaneously:

- Eq 1: $x + 4y = -3 \implies x = -4y - 3$
- Eq 2: $4x + 9y = -5$

Substitute x from Eq 1 into Eq 2:

$$4(-4y - 3) + 9y = -5$$

$$-16y - 12 + 9y = -5 \implies -7y = 7 \implies \mathbf{\bar{y} = -1}$$

Substitute $y = -1$ back into Eq 1:

$$x + 4(-1) = -3 \implies x - 4 = -3 \implies \mathbf{\bar{x} = 1}$$

2. Correlation coefficient (r)

We must correctly identify which equation is the line of Y on X and which is X on Y. We test this by ensuring the product of regression coefficients $r^2 \leq 1$.

Variable	Regression Line Assignment	Coefficients Derived	Validity (r2=byx×bxy
Test 1	Eq 1 is X on Y: $x = -4y - 3$	$b_{xy} = -4$	$r^2 = (-4) \times (-\frac{4}{9}) = \frac{16}{9}$
	Eq 2 is Y on X: $9y = -4x - 5 \implies y = -\frac{4}{9}x - \frac{5}{9}$	$b_{yx} = -\frac{4}{9}$	Invalid (> 1)
Test 2	Eq 1 is Y on X: $4y = -x - 3 \implies y = -\frac{1}{4}x - \frac{3}{4}$	$b_{yx} = -\frac{1}{4}$	$r^2 = (-\frac{1}{4}) \times (-\frac{9}{4}) = \frac{9}{1}$
	Eq 2 is X on Y: $4x = -9y - 5 \implies x = -\frac{9}{4}y - \frac{5}{4}$	$b_{xy} = -\frac{9}{4}$	Valid (≤ 1)

Since the valid regression coefficients (b_{yx} and b_{xy}) are both negative, the correlation coefficient r must also be negative.

$$r = -\sqrt{\frac{9}{16}} = -\frac{3}{4} = \mathbf{-0.75}$$

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b. The first three moments about the value 7 from 9 observations are 0.2, 19.4 and -41.0. Find the measures of central tendency and dispersion. (3 marks)

Let the arbitrary origin be $A = 7$. We are given the sample moments about A:

- First moment about A: $m'_1 = 0.2$
- Second moment about A: $m'_2 = 19.4$
- Third moment about A: $m'_3 = -41.0$

1. Measure of Central Tendency (Mean)

The mean ($\bar{x}$) is calculated by adding the arbitrary origin to the first moment about that origin:

$$\bar{x} = A + m'_1$$

$$\bar{x} = 7 + 0.2 = \mathbf{7.2}$$

2. Measure of Dispersion (Variance & Standard Deviation)
The primary measure of dispersion is the Variance (m_2, or the second central moment), which leads to Standard Deviation. We use the relationship between moments about an arbitrary origin and moments about the mean:

$$Variance(m_2) = m'_2 - (m'_1)^2$$

$$m_2 = 19.4 - (0.2)^2$$

$$m_2 = 19.4 - 0.04 = \mathbf{19.36}$$

Standard Deviation (σ) = $\sqrt{19.36} = \mathbf{4.4}$.
(Note: While the 3rd moment (m'_3) is provided, it is used to find skewness, which describes shape/symmetry rather than central tendency or dispersion. The required answers are Mean = 7.2 and Variance = 19.36).

Set 3

CO1

a. Define compound probability of two events. (2 marks)
The compound probability (also known as joint probability) of two events A and B is the probability that both events occur simultaneously or in succession. It is denoted by $P(A \cap B)$.
According to the multiplication rule of probability:

- If A and B are dependent events, $P(A \cap B) = P(A) \cdot P(B|A)$
- If A and B are independent events, $P(A \cap B) = P(A) \cdot P(B)$

b. The contents of urn I, II and III are as follows: 1 W, 2 B, 3 R balls; 2 W, 1 B, 1 R balls; 4 W, 5 B, 3 R balls. One urn is chosen at random and two balls are drawn. They happen to be W and R. What is probability that they come from urn II? (5 marks)

Let's define our events:

- E_1, E_2, E_3: Choosing Urn I, II, and III respectively. Since chosen at random, $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$.
- A: Drawing exactly 1 White (W) and 1 Red (R) ball.

We need to find the conditional probabilities of drawing 1W, 1R from each urn:

- **Urn I (1W, 2B, 3R | Total = 6):**

$$P(A|E_1) = \frac{\binom{1}{1} \times \binom{3}{1}}{\binom{6}{2}} = \frac{1 \times 3}{15} = \frac{1}{5}$$

- **Urn II (2W, 1B, 1R | Total = 4):**

$$P(A|E_2) = \frac{\binom{2}{1} \times \binom{1}{1}}{\binom{4}{2}} = \frac{2 \times 1}{6} = \frac{1}{3}$$

- **Urn III (4W, 5B, 3R | Total = 12):**

$$P(A|E_3) = \frac{\binom{4}{1} \times \binom{3}{1}}{\binom{12}{2}} = \frac{4 \times 3}{66} = \frac{12}{66} = \frac{2}{11}$$

We must find $P(E_2|A)$ using Bayes' Theorem:

$$P(E_2|A) = \frac{P(E_2)P(A|E_2)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + P(E_3)P(A|E_3)}$$

Since $P(E_i) = \frac{1}{3}$ for all urns, we can cancel it out from the numerator and denominator:

$$P(E_2|A) = \frac{\frac{1}{3}}{\frac{1}{5} + \frac{1}{3} + \frac{2}{11}}$$

$$P(E_2|A) = \frac{\frac{1}{3}}{\frac{33+55+30}{165}} = \frac{\frac{1}{3}}{\frac{118}{165}}$$

$$P(E_2|A) = \frac{1}{3} \times \frac{165}{118} = \frac{55}{118}$$

Answer: The probability that the balls came from Urn II is $\frac{55}{118}$.

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c. For what value of "a" will the function $f(x) = ax; x = 1, 2, 3, \dots n$ be the pmf of a discrete random variable X. Find the mean and variance of X. (3 marks)

1. Finding "a":

For $f(x)$ to be a valid Probability Mass Function (PMF), the sum of all probabilities must equal 1.

$$\sum_{x=1}^{n} f(x) = 1 \implies \sum_{x=1}^{n} ax = 1$$

$$a\sum_{x=1}^{n} x = 1 \implies a\left(\frac{n(n+1)}{2}\right) = 1$$

$$\mathbf{a = \frac{2}{n(n+1)}}$$

2. Finding the Mean $E(X)$:

$$E(X) = \sum_{x=1}^{n} x \cdot f(x) = \sum_{x=1}^{n} x(ax) = a \sum_{x=1}^{n} x^2$$

Using the sum of squares formula $\sum x^2 = \frac{n(n+1)(2n+1)}{6}$ and substituting a:

$$E(X) = \left(\frac{2}{n(n+1)}\right)\left(\frac{n(n+1)(2n+1)}{6}\right) = \frac{\mathbf{2n+1}}{\mathbf{3}}$$

3. Finding the Variance $Var(X)$:

First, find $E(X^2)$:

$$E(X^2) = \sum_{x=1}^{n} x^2 \cdot f(x) = \sum_{x=1}^{n} x^2(ax) = a \sum_{x=1}^{n} x^3$$

Using the sum of cubes formula $\sum x^3 = \left(\frac{n(n+1)}{2}\right)^2$:

$$E(X^2) = \left(\frac{2}{n(n+1)}\right)\left(\frac{n(n+1)}{2}\right)^2 = \frac{n(n+1)}{2}$$

Now, $Var(X) = E(X^2) - [E(X)]^2$:

$$Var(X) = \frac{n(n+1)}{2} - \left(\frac{2n+1}{3}\right)^2$$

$$Var(X) = \frac{n^2+n}{2} - \frac{4n^2+4n+1}{9}$$

Getting a common denominator of 18:

$$Var(X) = \frac{9(n^2+n) - 2(4n^2+4n+1)}{18} = \frac{9n^2+9n-8n^2-8n-2}{18}$$

$$\mathbf{Var(X) = \frac{n^2+n-2}{18}}$$

(Note: This factors to $\frac{(n+2)(n-1)}{18}$)

CO2

a. Approximate i) Poisson distribution from Binomial distribution and ii) Normal distribution from Poisson distribution. (3+3 marks)

i) Poisson approximation from Binomial distribution:

A Binomial distribution with parameters n (number of trials) and p (probability of success) can be approximated by a Poisson distribution under the following limiting conditions:

1. The number of trials is indefinitely large ($n \to \infty$).
2. The probability of success for each trial is indefinitely small ($p \to 0$).
3. The mean, $np = \lambda$, is a finite constant.
   Under these conditions, the Binomial PMF converges to the Poisson PMF:

$$\lim_{n\to\infty, p\to 0} \binom{n}{x} p^x (1-p)^{n-x} = \frac{e^{-\lambda}\lambda^x}{x!}$$

ii) Normal approximation from Poisson distribution:

A Poisson distribution with parameter λ can be approximated by a Normal distribution under the limiting condition where:

1. The parameter λ (which is both the mean and variance) becomes indefinitely large ($\lambda \to \infty$).
   Under this condition, the Poisson variable X can be standardized as $Z = \frac{X-\lambda}{\sqrt{\lambda}}$. As $\lambda \to \infty$, the distribution of Z approaches the Standard Normal distribution $N(0,1)$. Thus, $X \sim N(\mu = \lambda, \sigma^2 = \lambda)$.

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b. A discrete random variable X follows Poisson distribution such that P(X = 1) = P(X = 2). Find the mean and variance of the distribution. (3 marks)

The PMF of a Poisson distribution is $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$.

Given $P(X = 1) = P(X = 2)$:

$$\frac{e^{-\lambda}\lambda^1}{1!} = \frac{e^{-\lambda}\lambda^2}{2!}$$

Since $e^{-\lambda}$ is never zero, we can divide both sides by $e^{-\lambda}$:

$$\lambda = \frac{\lambda^2}{2} \implies \lambda^2 - 2\lambda = 0 \implies \lambda(\lambda - 2) = 0$$

Since λ must be strictly greater than 0 for a Poisson distribution, we have $\lambda = 2$.

In a Poisson distribution, both Mean and Variance are equal to the parameter λ.

Answer: Mean = **2**, Variance = **2**.

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c. In an examination marks obtained by the student in C, JAVA and Python are independently and normally distributed with mean 50, 52, 48 and s.ds. 15, 12, 16 respectively. Find the probability that total marks of a student are between 90 and 180 (Given $\phi(1.2) = 0.8849$ and $\phi(2.4) = 0.9918$). (3 marks)

Let the marks be X_C, X_J, and X_P. We are given:

- $X_C \sim N(50, 15^2)$
- $X_J \sim N(52, 12^2)$
- $X_P \sim N(48, 16^2)$

Let T be the total marks: $T = X_C + X_J + X_P$.

Since they are independent normal variables, T is also normally distributed.

- **Mean of T (μ_T):** $\mu_T = 50 + 52 + 48 = 150$

- **Variance of T (σ_T^2):** $\sigma_T^2 = 15^2 + 12^2 + 16^2 = 225 + 144 + 256 = 625$
- **Standard Deviation (σ_T):** $\sigma_T = \sqrt{625} = 25$

We need to find $P(90 < T < 180)$. We standardize to $Z = \frac{T-\mu_T}{\sigma_T}$:

- For $T = 90$: $Z_1 = \frac{90-150}{25} = \frac{-60}{25} = -2.4$
- For $T = 180$: $Z_2 = \frac{180-150}{25} = \frac{30}{25} = 1.2$

$$P(90 < T < 180) = P(-2.4 < Z < 1.2)$$

$$= P(Z < 1.2) - P(Z < -2.4)$$

Using the symmetry of the normal curve:
$P(Z < -2.4) = 1 - P(Z < 2.4) = 1 - \phi(2.4)$.

$$P = \phi(1.2) - [1 - \phi(2.4)]$$

$$P = 0.8849 - [1 - 0.9918]$$

$$P = 0.8849 - 0.0082 = \mathbf{0.8767}$$

CO3

a. The regression equation calculated from a given set of observations are $x = -0.2y + 4.2$ and $y = -0.8x + 8.4$. Calculate r and estimated value of y when x = 4. (5 marks)

1. Calculate correlation coefficient (r):

From the equations, we identify the regression coefficients.

- Eq 1 is x on y: $x = -0.2y + 4.2 \implies b_{xy} = -0.2$
- Eq 2 is y on x: $y = -0.8x + 8.4 \implies b_{yx} = -0.8$

We verify $r^2 = b_{xy} \times b_{yx} = (-0.2) \times (-0.8) = 0.16$. Since $0.16 \leq 1$, this assumption is valid.

Since both regression coefficients are negative, r must also be negative.

$$r = -\sqrt{0.16} = \mathbf{-0.4}$$

2. Estimated value of y when x = 4:

To estimate y given an x value, we use the regression equation of y on x (Eq 2).

$$y = -0.8x + 8.4$$

Substitute $x = 4$:

$$y = -0.8(4) + 8.4 = -3.2 + 8.4 = \mathbf{5.2}$$

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b. The first two moments of a distribution about 4 are -1.5 and 2.7. Calculate the mean and s.d. (3 marks)

Let the arbitrary origin be $A = 4$. We are given moments about A:

- First moment (m'_1): -1.5
- Second moment (m'_2): 2.7

1. Calculate Mean ($\bar{x}$):

The mean is the origin plus the first moment about that origin.

$$\bar{x} = A + m'_1$$

$$\bar{x} = 4 + (-1.5) = \mathbf{2.5}$$

2. Calculate Standard Deviation (s.d.):

First, find the variance (m_2), which is the second central moment.

$$Variance(m_2) = m'_2 - (m'_1)^2$$

$$m_2 = 2.7 - (-1.5)^2$$

$$m_2 = 2.7 - 2.25 = 0.45$$

Standard Deviation is the square root of the variance:

$$s.d.(\sigma) = \sqrt{0.45} \approx \mathbf{0.6708}$$